

## # Temp Variation of carrier concentration in Extrinsic S.C

for n-type

- Let  $N_d$  be the concentration of donor in the material. At  $0^\circ\text{K}$  the donor atoms are not ionized and are the level  $E_d$ .
- At temperature above  $0^\circ\text{K}$ , the donor atoms get ionized and free electrons jump to conduction band.
- Assume that the electron concentration,  $n$  in the conduction band is equal to the number of donor i.e.  $n = N_d^+$

where,  $n = 2 \left[ \frac{2\pi m_e^* kT}{h^2} \right]^{3/2} \exp\left(\frac{E_f - E_c}{kT}\right)$  — (1)

Let  $N_c = 2 \left[ \frac{2\pi m_e^* kT}{h^2} \right]^{3/2}$

Therefore  $n = N_c \exp\left(\frac{E_f - E_c}{kT}\right)$  — (2)

$N_d^+ = N_d \exp\left(\frac{E_d - E_f}{kT}\right)$  — (3)

Equate the above eq<sup>n</sup> i.e.  $n = N_d^+$

$N_c \exp\left(\frac{E_f - E_c}{kT}\right) = N_d \exp\left(\frac{E_d - E_f}{kT}\right)$  — (4) Energy level diagram of n-type S.C.

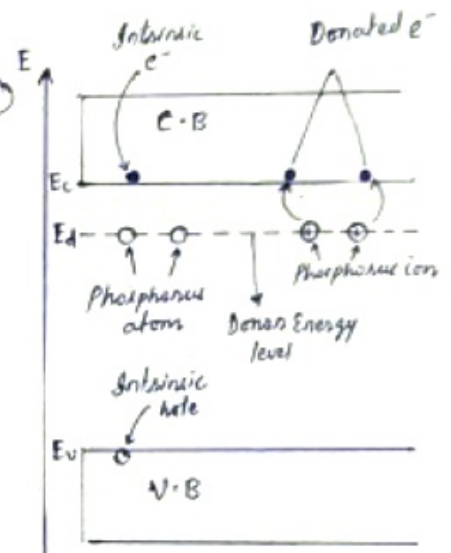
Taking logarithm and rearranging

$\left(\frac{E_d - E_f}{kT}\right) + \left(\frac{E_c - E_f}{kT}\right) = \ln\left(\frac{N_c}{N_d}\right)$  — (5)

$(E_d + E_c) - 2E_f = kT \ln\left(\frac{N_c}{N_d}\right)$  — (6)

$E_f = \left(\frac{E_d + E_c}{2}\right) - \left(\frac{kT}{2}\right) \ln\left(\frac{N_c}{N_d}\right)$  — (7)

$E_f = \left(\frac{E_d + E_c}{2}\right) + \left(\frac{kT}{2}\right) \ln\left(\frac{N_d}{N_c}\right)$  — (8)

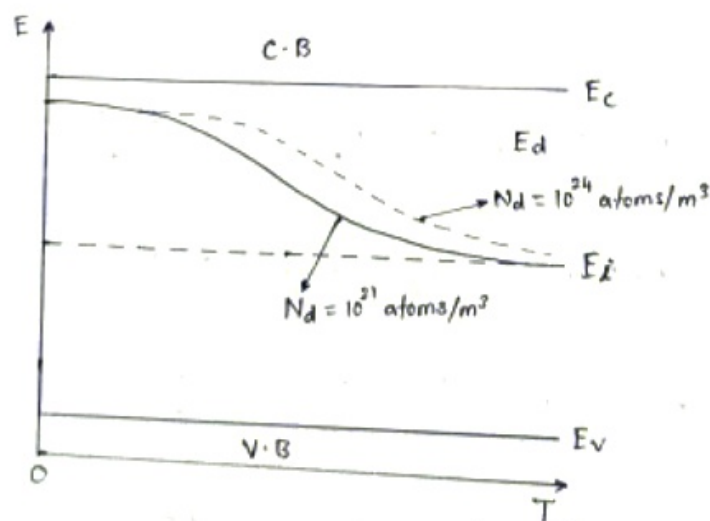


At  $T = 0^\circ\text{K}$ ,  $E_f = \left(\frac{E_d + E_c}{2}\right)$  — (9)

Substitute the value of  $E_f$  from eq<sup>n</sup> (1) in eq<sup>n</sup> (8) and rearrange.

$$n = (2N_d)^{1/2} \left(\frac{2\pi m_e^* kT}{h^2}\right)^{3/4} \exp\left(\frac{E_d - E_c}{2kT}\right) \text{ — (10)}$$

- Hence the density of the electrons in the C.B is proportional to the square root of the donor concentration at low temp.
- At higher temp intrinsic behaviour predominates and donor concentration becomes insignificant.
- With an increase of temp  $E_f$  increases slightly as per eq<sup>n</sup> (8)
- As the temp is further increased more no of donor atoms are ionized and at a particular temp all the donor atoms are ionized.
- Further increase in temp results in generation of electron-hole pairs due to breaking of covalent bonds and the material tends to behave in intrinsic manner.



Variation of Fermi level with  $T$  and  $N_d$

# # Temp Variation of carrier concentration in Extrinsic S.C

for P-type

- Let  $N_a$  be the concentration of acceptor in the material.  
At 0°K, the acceptor atoms are not ionized and are at the level  $E_a$ .
- At temp above 0°K, the acceptor atoms get ionized and holes appear in the valence band.
- Assume that the hole concentration,  $P$  in the valence band is equal to the number of acceptor. i.e  $P = N_a^-$  where

$$P = 2 \left[ \frac{2\pi m_h^* kT}{h^2} \right]^{3/2} \exp\left(\frac{E_v - E_f}{kT}\right) \quad \text{--- (1)}$$

$$\text{Let } N_v = 2 \left[ \frac{2\pi m_h^* kT}{h^2} \right]^{3/2}$$

$$\text{Therefore } P = N_v \exp\left(\frac{E_v - E_f}{kT}\right) \quad \text{--- (2)}$$

$$N_a^- = N_a \exp\left(\frac{E_f - E_a}{kT}\right) \quad \text{--- (3)}$$

Equate the above equations i.e  $P = N_a^-$

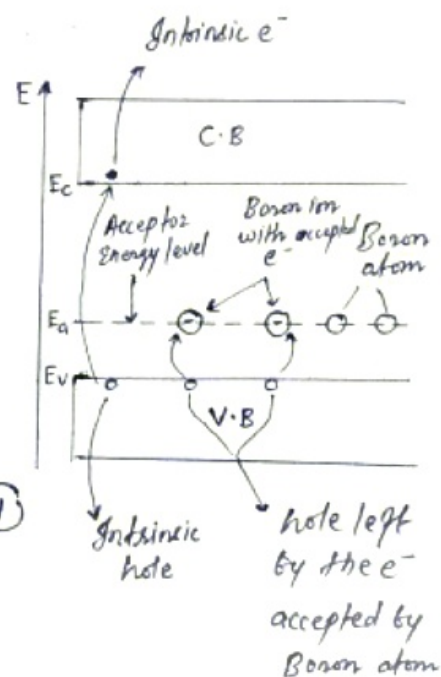
$$N_v \exp\left(\frac{E_v - E_f}{kT}\right) = N_a \exp\left(\frac{E_f - E_a}{kT}\right) \quad \text{--- (4)}$$

Taking logarithm and rearranging.

$$\left(\frac{E_f - E_a}{kT}\right) - \left(\frac{E_v - E_f}{kT}\right) = \ln\left(\frac{N_v}{N_a}\right) \quad \text{--- (5)}$$

$$-(E_v + E_a) + 2E_f = (kT) \ln\left(\frac{N_v}{N_a}\right) \quad \text{--- (6)}$$

$$E_f = \left(\frac{E_v + E_a}{2}\right) + \left(\frac{kT}{2}\right) \ln\left(\frac{N_v}{N_a}\right) \quad \text{--- (7)}$$



Energy band diagram for P-type S.C

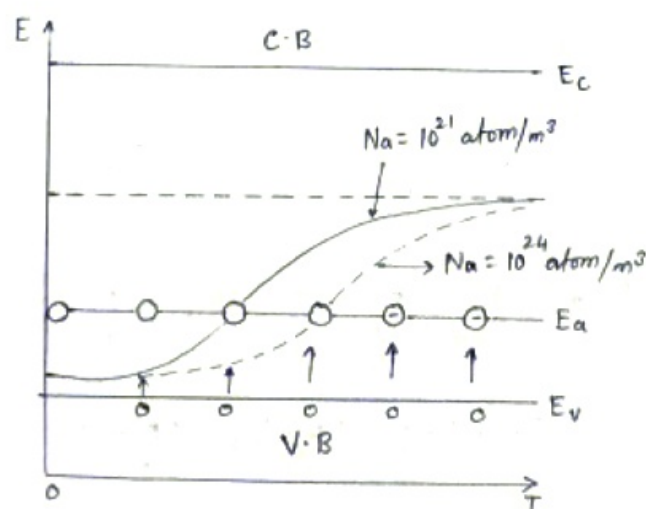
$$E_f = \left( \frac{E_v + E_a}{2} \right) - \left( \frac{kT}{2} \right) \ln \left( \frac{N_a}{N_v} \right) \quad \text{--- (8)}$$

$$\text{At } T=0^\circ\text{K}, E_f = \left( \frac{E_v + E_a}{2} \right) \quad \text{--- (9)}$$

Substitute the value of  $E_f$  from eq<sup>n</sup> (8) in eq<sup>n</sup> (1) and rearrange.

$$P = (2N_a)^{1/2} \left( \frac{2\pi m_h^* kT}{h^2} \right)^{1/2} \exp \left( \frac{E_v - E_a}{2kT} \right) \quad \text{--- (10)}$$

- Hence the density of holes in the valence band is proportional to the square root of the acceptor concentration at low temp.
- At higher temperature intrinsic behaviour predominates and acceptor concentration becomes insignificant.
- With an increase of temp  $E_f$  decreases slightly as per eq<sup>n</sup> (8)
- As temp increases Fermi level decreases since more acceptor atoms are ionized. At a particular temp all the acceptor atoms are ionized.
- Further increase in temp results in generation of  $e^-$  - hole pairs due to breaking of covalent bonds. The material tends to behave in intrinsic manner.



Variation of Fermi level with  $T$  in P-type s.c